Tour de Maths 2024

Leg 3

Question Paper

With special thanks to the sponsorship of

π = 3, 14159 26535 89793 23846 26433 83279 50288 41971 69399 37510 58209 74944 59230 78164 06286 20899 86280 34825 34211 70679 82148 08651 32823 06647 09384 46095 50582 23172 53594 08128 48111 74502 84102 70193 85211 05559 64462 29489 54930 38196 44288 10975 66593 34461 28475 64823 37867 83165 27120 19091 45648 56692 34603 48610 45432 66482 13393 60726 02491 41273 72458 70066 06315 58817 48815 20920 96282 92540 91715 36436 78925 90360 01133 05305 48820 46652 13841 46951 94151 16094 33057 27036 57595 91953 09218 61173 ...

Multiple choice section. 10 questions, 5 marks each.

QUESTION 1

What is the probability that a 3 will occur within the first 50 decimal places of π. This excludes the leading 3.

A: 10% B: 14% C: 16% D: 18%

QUESTION 2

Some students use $\pi = \frac{22}{7}$ in their calculations.

As a percentage, how accurate is this estimation?

A: 4% B: 0,4% C: 0,04% D: 0,004%

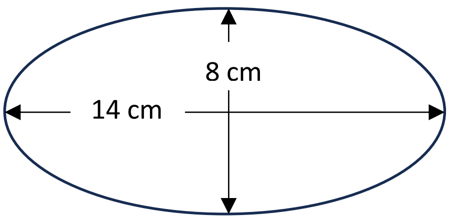
QUESTION 3

The area of an ellipse is given as $A = \pi ab$ where height is 2a and length is 2b.

In this ellipse the height is 8 cm and the length 14 cm.

Calculate the area of the ellipse alongside

In terms of π.



A: 22π B: 28π C: 30π D: 36π

QUESTION 4

At midnight a frog falls into a hole that is 14,5 meters deep. It can jump 2m upwards but slides back half a meter each time as it rests before the next jump.

How many jumps does it take to get out of the hole?

A: 10 jumps B: 11 jumps C: 12 jumps D: 13 jumps

QUESTION 5

An abundant number is defined as a number where the sum of all the positive **proper** factors is greater than the number itself. The proper factors exclude the number and one.

Example:

$F_{22} = 1; 2; 11; 22$ The sum of proper factors: $2 + 11 = 13.$ 22 is not abundant.

$F_{12} = 1; 2; 3; 4; 6; 12$ The sum of proper factors: $2 + 3 + 4 + 6 = 15$ 12 is abundant.

What is the smallest abundant number greater than 24?

A: 26 B: 28 C: 30 D: 36

QUESTION 6

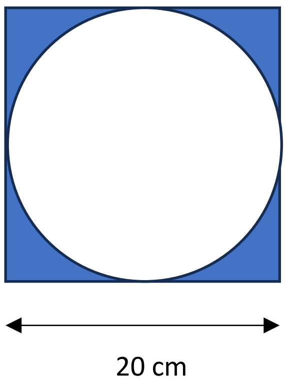
A six-digit number is written as: 839A6B where A and B are unknown digits.

The number is divisible by 11 and 4. What is the greatest possible value of A+B?

A: 8 B: 9 C: 10 D: 11

QUESTION 7

The figure is a circle inscribed inside a square. Give the area of the shaded region correct to two decimal places.



A: 40,84 B: 75,00 C: 80,12 D: 85,84

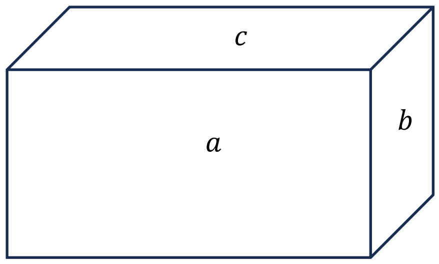
QUESTION 8

How many non-congruent right angles triangles with integer sides exist, where the perimeter is less than 75.

A: 7 B: 9 C: 11 D: 14

QUESTION 9

Three faces of a prism have areas a, b, and c as shown. What is the volume of the box?



A: $2(a+b+c)$ B: $\sqrt{abc}$ C: $\sqrt{a^2+b^2+c^2}$ D: $(a+b+c)$

QUESTION 10

What is the last digit (the units digit) of: $2^{2024} \cdot 3^{2024} \cdot 5^{2024}$?

A: 0 B: 2 C: 5 D: 7

These questions require an answer.

Simplest surd form or answers in terms of pi are best. Decimal answers will also be accepted.

10 mark questions.

QUESTION 11

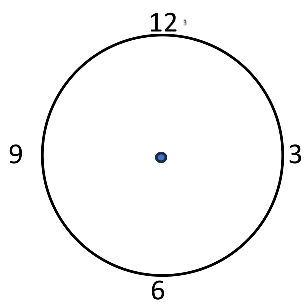
Decimal Time: Decimal time is convenient for mathematical calculations, but often needs to be converted back to hours and minutes for everyday usage. Decimal time format is also used to compare time durations in a decimal number.

For example, 12:30:15 is represented as 12,5025. One hour is equal to 1,0 and one decimal minute is equal to 1/60 and one second is equal to 1/3600.

An event started at 10 minutes past 5 in the evening. What is this in decimal time?

QUESTION 12

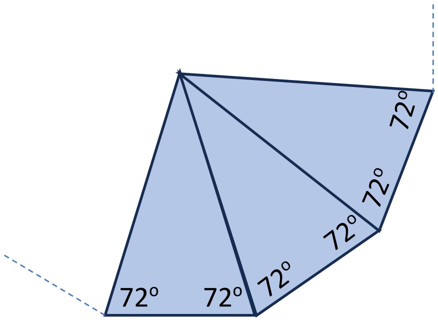
At half-past three, what is the angles formed between the minute hand and the hour hand of a clock?



QUESTION 13

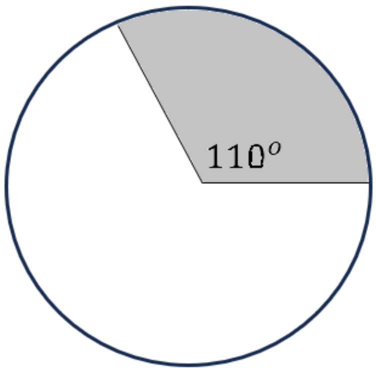
A series of isosceles triangles all with the equal angles of 72° (as shown) are placed side by side. More triangles are added to make a complete polygon with no overlaps.

What is the number of sides of the polygon?



QUESTION 14

Find the radius of the circle if the area of the shaded region is 50π units².



QUESTION 15

Tetration is a cool way of expression numbers in a 'power tower'. ${}^{3}5 = 5^{5^5}$.

Meaning 5 to the power 5 to the power 5.

What is the value of ${}^{3}3 \times {}^{4}2$? Leave your answer in exponential form.

20 mark questions

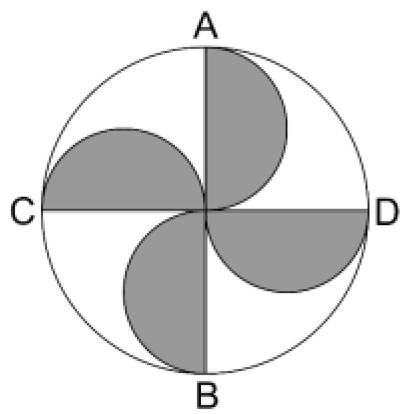
QUESTION 16

There is a total of 45 boys and girls in a choir. The mean age of the 18 boys is 16,2 years. The mean age of the 27 girls is 16,7 years.

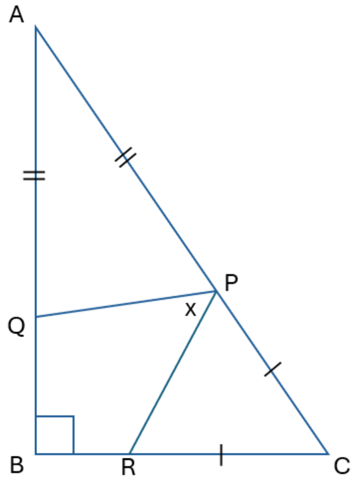
Calculate the mean age of all 45 boys and girls.

QUESTION 17

In the circle shown, diameters AB and CD are perpendicular, and each of the four shaded regions is a semicircle. If the ratio of the unshaded area to the shaded area is $\frac{u}{s}$ in simplest form, then what is the value of $u + s$?



QUESTION 18



ABC is a right-angled triangle, and points P,Q, and R are chosen on the sides so that

AP = AQ and CP = CR. The diagram is not drawn to scale.

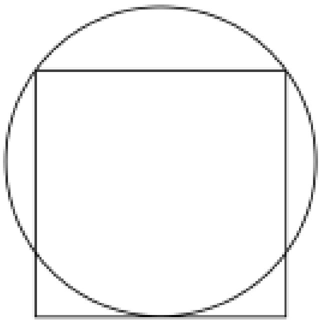
What is the value of x in degrees?

QUESTION 19

A satellite orbits the earth 400km above the earth at a speed of 28 000 km/h. The earth is assumed to be spherical with a circumference of 40 075 km.

How long is one orbit in minutes?

QUESTION 20



In the figure the square with side length 8 has two vertices on the circle, and one side touching the circle.

What is the length of the radius of the circle?

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